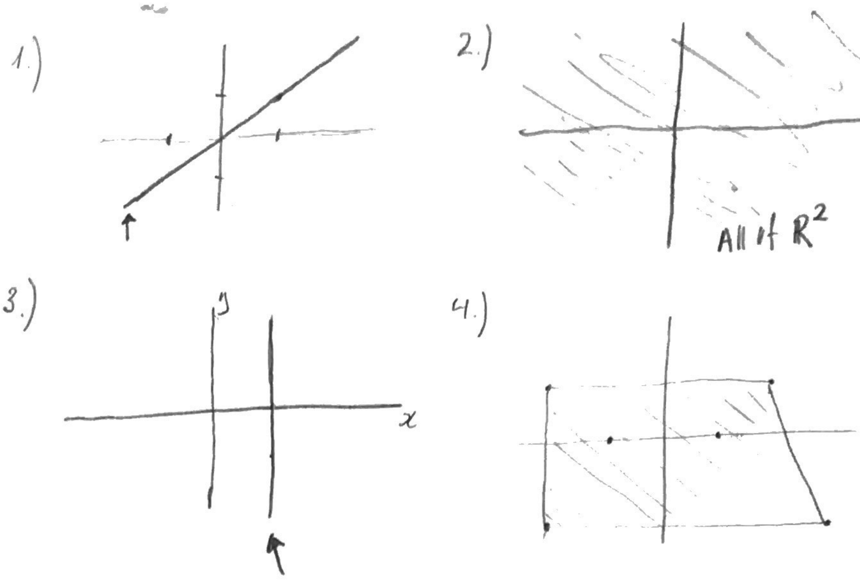


Convex Sets (1 pt)

Example sets

Sketch the following sets in $\mathbb{R}^2$

1. $\operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}-0.5\\-0.5\end{pmatrix}\right\}$
2. $\operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}0.5\\-0.5\end{pmatrix}\right\}$
3. $\operatorname{aff}\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}1\\-1\end{pmatrix}\right\}$
4. $\operatorname{conv}\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}2\\1\end{pmatrix},\begin{pmatrix}3\\-2\end{pmatrix},\begin{pmatrix}-1\\0\end{pmatrix},\begin{pmatrix}-2\\1\end{pmatrix},\begin{pmatrix}-2\\-2\end{pmatrix}\right\}$

Convexity

Let $C \subset \mathbb{R}^n$ be a convex set, with $x_1,\ldots,x_k \in C$, and let $\theta_1,\ldots,\theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0, \theta_1 + \ldots + \theta_k = 1$. Show that $\theta_1 x_1 + \ldots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k.) Hint. Use induction on k.

Base. Let $k=2$: $\theta_1 x_1 + \theta_2 x_2$. $\theta_2 = 1-\theta_1$ since $\theta_1+\theta_2=1$. $\Rightarrow \theta_1 x_1 + (1-\theta_1)x_2 \in C$ (by def. of convexity)

Induction: Assume $P_k = \theta_1 x_1 + \ldots + \theta_k x_k \in C$. For $P_{k+1} = \theta_1 x_1 + \ldots + \theta_k x_k + \theta_{k+1}x_{k+1}$, let $\theta_{k+1} = 0$, then $P_{k+1} = P_k \in C$ ■.

Linear Equations

Show that the solution set of linear equations $\{x | Ax = b\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ is an affine set.

Let $S = \{x \mid Ax = b\}$ (i.e. $Ax_1 = b$ and $Ax_2 = b$)

Let $x_1, x_2 \in S$. Want to show that $\alpha_1 x_1 + \alpha_2 x_2 \in S$, i.e. $A(\alpha x_1 + \alpha_2 x_2) = b$) with: $\alpha_1 + \alpha_2 = 1$ $\Rightarrow \alpha_2 = 1 - \alpha_1$

$A(\alpha_1 x_1 + (1-\alpha_1)x_2) = \alpha_1 A x_1 + (1-\alpha_1)Ax_2 = \alpha_1 b + (1-\alpha_1) b = b$ ■

Linear Inequations

1. Show that the solution set of linear inequations $\{x | Ax \preceq b, Cx = d\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times n}$ and $d \in \mathbb{R}^k$ is a convex set. Here $\preceq$ means componentwise less or equal.

Let $S = \{x \mid Ax \preceq b, Cx = d\}$ and $x_1, x_2 \in S$. Want to show $\alpha_1 x_1 + \alpha_2 x_2 \in S$, where $\alpha_1 + \alpha_2 = 1$, $\alpha_{1,2} \geq 0$ $\Rightarrow \alpha_2 = 1-\alpha_1$, $0 \leq \alpha_1 \leq 1$

$A(\alpha_1 x_1 + (1-\alpha_1)x_2) = \alpha_1 A x_1 + (1-\alpha_1)Ax_2 \preceq \alpha_1 b + (1-\alpha_1) b = b$ (since $\alpha_1, 1-\alpha_1 \geq 0$)

$C(\alpha_1 x_1 + (1-\alpha_1)x_2) = \alpha_1 C x_1 + (1-\alpha_1) C x_2 = \alpha_1 d + (1-\alpha_1) d = d$ ■

2. Is it an affine set?

$C = 0_{1\times1}$ $d = [0]$

NO. Let $x_1 = [0]$, $x_2 = [1]$, $A = I_{1\times1}$, $b = [1]$. Clearly, $x_1, x_2 \in S$ (as defined in part 1.) (note; $x_1, x_2 \in \mathbb{R}^1$)

Let $\alpha_1 = -1$, $\alpha_2 = 2$: $A(-1[0] + 2[1]) = [2] > b$, hence $(\alpha_1 x_1 + \alpha_2 x_2) \notin S$ ∎

$(\alpha_1 + \alpha_2 = 1 \Rightarrow$ affine$)$

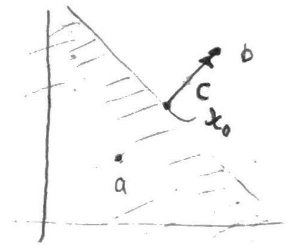
↱ either half of the space divided by a (hyper) plane.

Voronoi description of halfspace

Let a and b be distinct points in $\mathbb{R}^n$. Show that the set of all points that are closer (in Euclidean norm) to a than b, i.e., $\{x \mid \|x-a\|^2 \leq \|x-b\|^2\}$, is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

$\vec{c} : \frac{1}{2}(\vec{b} - \vec{a})$

$d : \vec{c}^T \vec{x}_0$



$\Rightarrow c^T x \leq d \Rightarrow c^T x - c^T x_0 \leq 0 \Rightarrow c^T(x - x_0) \leq 0 \Rightarrow \boxed{\frac{1}{2}(b^T - a^T)(x - x_0) \leq 0 \text{ where } x_0 = a + \frac{1}{2}(b-a)}$

Convex Illumination Problem (1 pt)

Show that the solution $p^* = (p_1^*, p_2^*, \ldots, p_n^*)^T \in \mathbb{R}^n$ of the non-convex illumination problem from the lecture

$$\begin{array}{ll} \text{minimize} & \max_{k=1\ldots m} |\log I_k - \log I_{des}| \\ \text{subject to} & 0 \leq p_j \leq p_{max}, \quad j = 1 \ldots n \end{array}$$

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with $I_k = \sum_{j=1}^{n} a_{kj} p_j$ for geometric constants $a_{jk} \in \mathbb{R}$, a constant desired illumination $I_{des} \in \mathbb{R}$ and an upper bound $p_{max} \in \mathbb{R}$ on the lamp power, is identical to the solution of the following equivalent (convex) problem

$$\begin{array}{ll} \text{minimize} & \max_{k=1\ldots m} h(I_k / I_{des}) \\ \text{subject to} & 0 \le p_j \le p_{max}, \quad j = 1\ldots n \end{array}$$

with $h(u) = \max\{u, 1/u\}$.

The objective function can be rewritten:

$$\max_{k=1\ldots m} |\log I_k - \log I_{des}| = \max_{k=1\ldots m} \left|\log \frac{I_k}{I_{des}}\right|$$

We define $u_k = \frac{I_k}{I_{des}}$. We know that $|\log u| = \max\{\log u, \underset{\uparrow}{-\log u}\}$ for $u > 0$.

$-\log u = \log \frac{1}{u}$.

$\log(x)$ is monotone increasing for $x \in (0, \infty)$

so $\max\{\log u, -\log u\} = \max\{\log u, \log \frac{1}{u}\} = \log \max\{u, \frac{1}{u}\} = \log h(u)$

Applying this to the objective:

$$\max_{k=1\ldots m} |\log I_k - \log I_{des}| = \max_{k=1\ldots m} \left|\log \frac{I_k}{I_{des}}\right| = \max_{k=1\ldots m} \log h(u) = \log \max_{k=1\ldots m} h(u)$$

As the log is monotone increasing, minimizing $\log \max_{k=1\ldots m} h(u)$ is the same as minimizing $\max_{k=1\ldots m} h\left(\frac{I_k}{I_{des}}\right)$ //